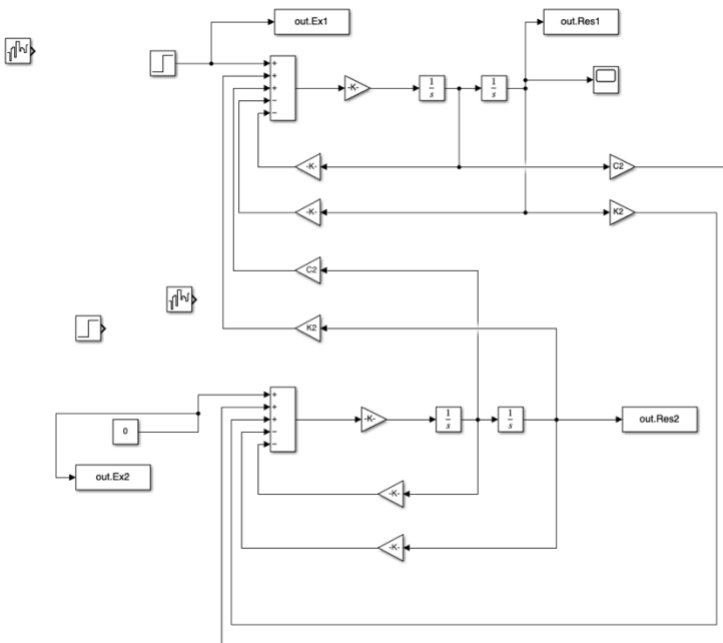


Report 8



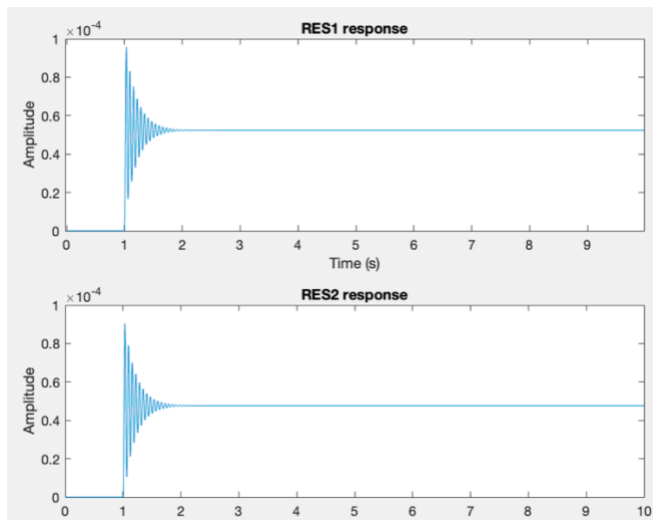
Substitute impulse forcing EX1, don't force EX2, collect responses from RES1 and RES2

RES1 frequency 1 damping: -0.6463

RES1 frequency 2 damping: -0.4361

RES2 frequency 1 damping: -0.6486

RES2 frequency 2 damping: -0.3505



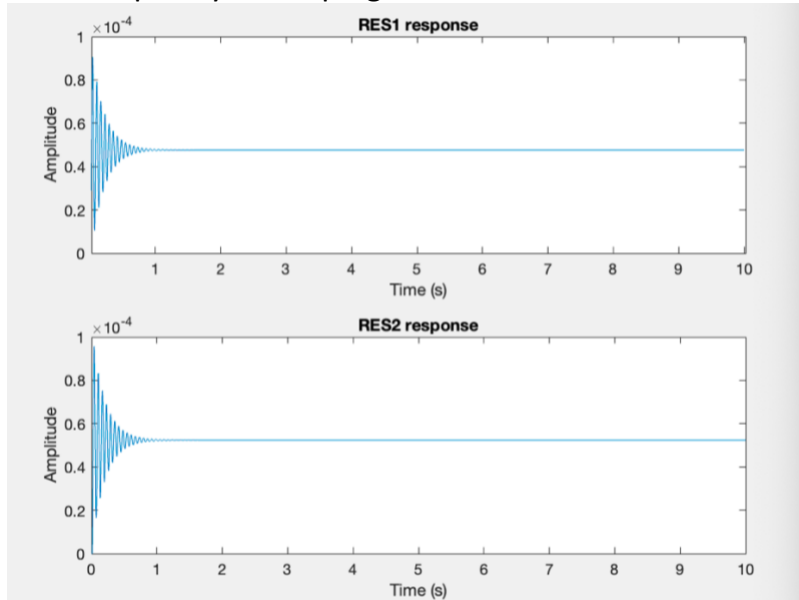
Substitute impulse forcing EX2, don't force EX1, collect responses from RES1 and RES2

RES1 frequency 1 damping: -0.8458

RES1 frequency 2 damping: -0.3623

RES2 frequency 1 damping: -0.8240

RES2 frequency 2 damping: -0.3635



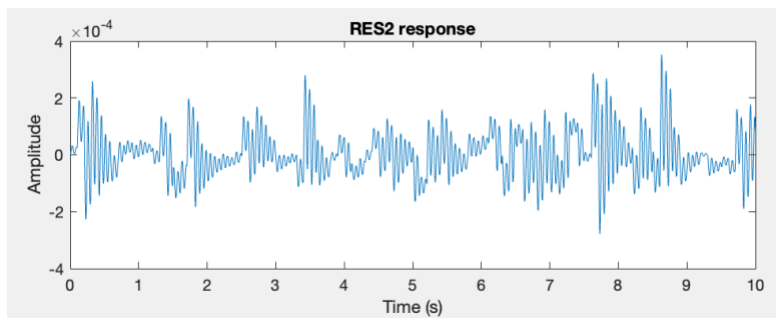
3) Substitute impulse forcing EX1, don't force EX2, collect responses from RES2. The total time should be 1 second with a sampling rate of 1000 Hz.

```
>> simumus
Warning: pk3 does not have at least 2 peaks.
> In simumus (line 53)
```

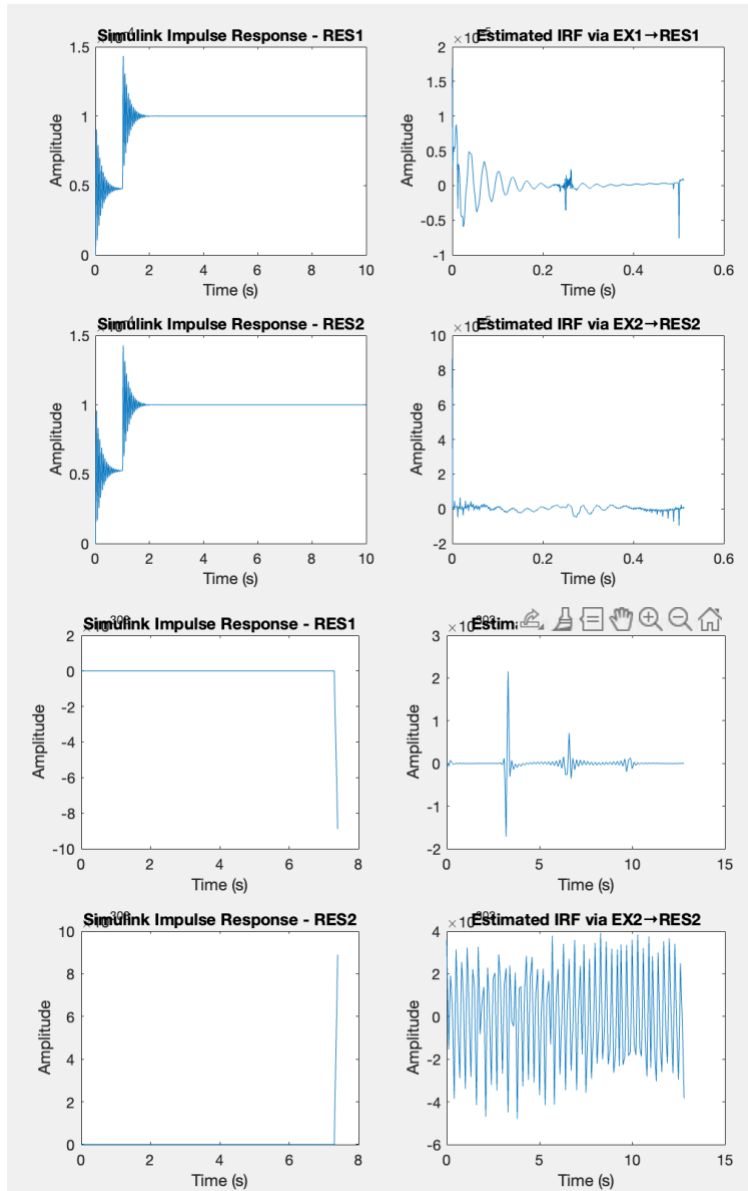
4) Use Band-Limited White Noise to force EX1, don't force EX2, collect responses from RES2. The total time should be 10 seconds with a sampling rate of 1000 Hz.

RES2 frequency 1 damping: 0.0212

RES2 frequency 2 damping: 0.1454

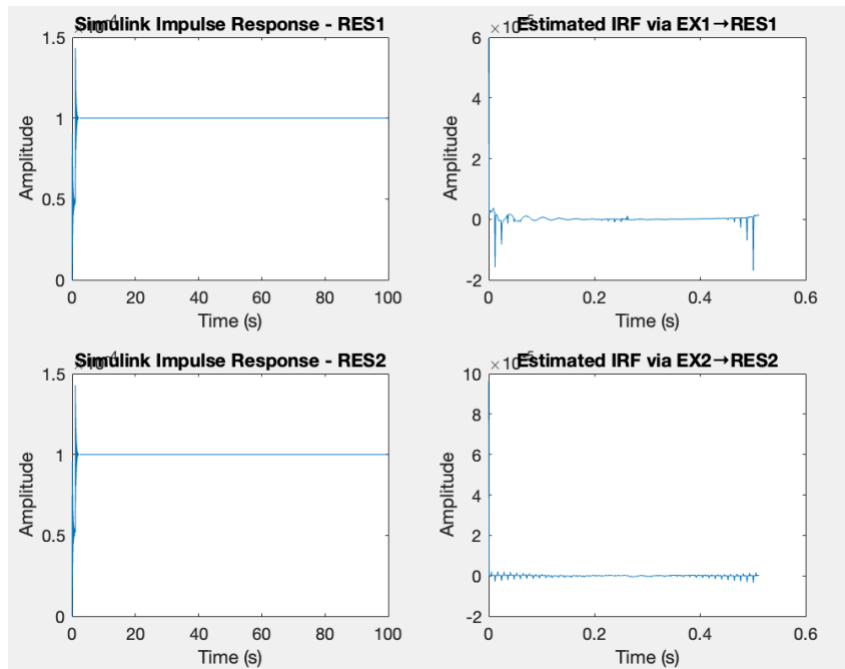


a) Estimate the spectral transition function from 4) using the "tfestimate" function, then determine the IRF impulse response function using the "ifft" function, compare the obtained impulse response function with the impulse response function from 3). For the estimation of the spectral transfer function, use the desired frequency resolution = 1 Hz.



b) Do 4) again with total time = 100 seconds, and do a) again with the desired frequency resolution

= 0.1 Hz. Again compare the obtained impulse response function 3) with the estimated impulse response function IRF.



c) Use the estimated spectral transfer functions (both i.e. a) and b)) to determine the attenuation coefficients using the half-power method. In addition, compare the obtained damping coefficients with the results of the estimation in the time domain.

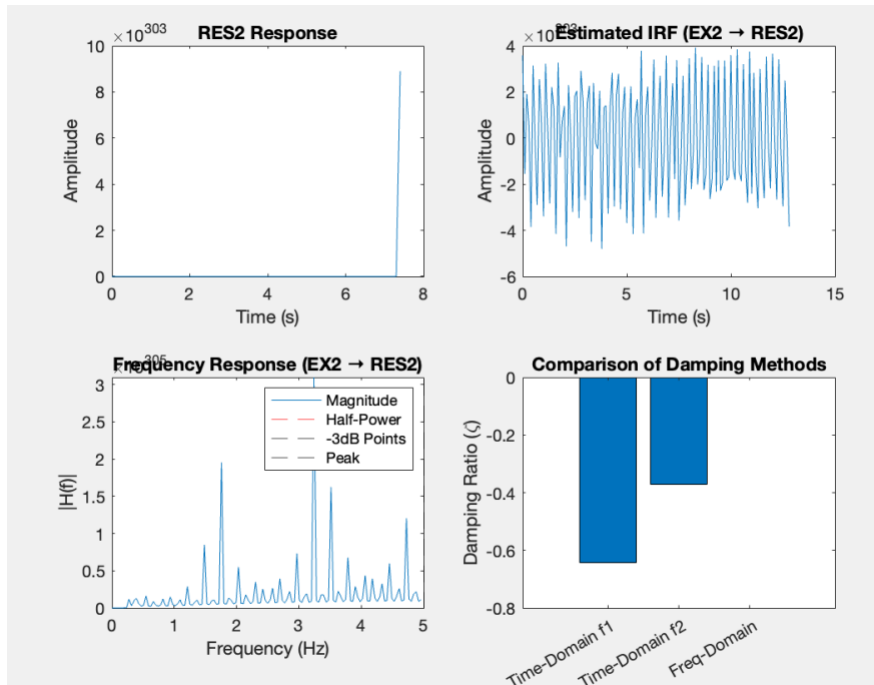
--- TIME DOMAIN DAMPING (log decrement) ---

RES2 frequency 1 (low-pass) damping: -0.6413

RES2 frequency 2 (high-pass) damping: -0.3704

--- FREQUENCY DOMAIN DAMPING (half-power) ---

Estimated zeta from half-power method: 0.0000



Based on the help function and information from lectures and the Internet, describe in your own words what the `modalfrf`, `modalsd` and `modalfit` functions are for. Also describe what a stabilization diagram is.

`modalfrf`

It's like checking how a structure responds when we shake it at different frequencies. For example, if you hit a metal beam and measure how it vibrates, `modalfrf` helps you understand at which frequencies it vibrates the most.

`modalsd`

This function finds possible natural frequencies and damping values of the system. But instead of just giving one solution, it tries different model complexities and plots all the results so we can see which modes are stable and which are not. It's like testing the system multiple times and seeing which results keep showing up.

`modalfit`

Imagine you've guessed the system's natural frequencies and damping, now you use `modalfit` to fine tune those guesses and create a mathematical model that matches your measured data as closely as possible.

Stabilization Diagram

Think of it as a chart that tells you which frequencies are real and stable because they appear consistently no matter how detailed your model is. If a mode appears over and over again at different model orders, it's probably a real physical mode of the structure, not just noise.